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ORIGINAL ARTICLE

Valaciclovir versus aciclovir for herpes simplex virus infection in HIV-infected individuals: two randomized trials

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Summary: Our objective was to evaluate valaciclovir for anogenital herpes in HIV-infected individuals using 2 controlled trials conducted before highly active antiretroviral therapy (HAART) was used. In Study 1, 1062 patients (CD4+ $\geqslant$100 cells/mm^3) received suppressive valaciclovir or aciclovir for one year and were assessed monthly. In Study 2, 467 patients were treated episodically for $\geqslant$5 days with valaciclovir or aciclovir and evaluated daily. Valaciclovir was as effective as aciclovir for suppression and episodic treatment of herpes. Hazard ratios [95% confidence interval (CI)] for time to recurrence for valaciclovir 500 mg twice daily and 1000 mg once daily *vs* aciclovir were 0.73 [0.50, 1.06], P=0.10, and 1.31 [0.94, 1.82], P=0.11. Valaciclovir 500 mg twice daily was superior to 1000 mg once daily, P=0.001. Valaciclovir 1000 mg twice daily was comparable to aciclovir on herpes episode duration (hazard ratio 0.92 [0.75, 1.14]). Adverse events were similar among treatments. In conclusion, valaciclovir is a safe, effective, convenient alternative to aciclovir for HSV infection in HIV-infected individuals.

Keywords: Valaciclovir, recurrent genital herpes, herpes simplex virus, HIV

INTRODUCTION

Recurrent anogenital lesions due to herpes simplex virus (HSV) reactivation are commonly described in individuals with human immunodeficiency virus (HIV) infection[1,2]. Seroprevalence for HSV in HIV-seropositive individuals is high, with as many as 80% of homosexual men having evidence of HSV infection[3–5]. As with many infections, the frequency and severity of HSV are heightened in patients who are immunocompromised compared with otherwise healthy immunocompetent subjects[2,6]. The clinical spectrum of disease in patients with HIV infection includes chronic genital or anogenital herpes, as well as severe recurrent facial and oral mucocutaneous HSV infection, oesophagitis, encephalitis and dissemination[6,7].

Reactivation of HSV may be associated with increased HIV viraemia, and genital ulceration is thought to facilitate acquisition and transmission of HIV infection[8–11]. Antiviral therapy to prevent or suppress recurrent HSV episodes produces considerable relief of symptoms, offers additional psychosocial benefits and the potential to influence favourably the natural history of HIV infection[12–15]. Suppressive aciclovir has commonly been effective in regimens of 200 mg 3 or 4 times daily or 400 mg twice daily[12,14]. Higher unit doses of 400 mg 4 or 5 times daily, however, have sometimes been necessary for effective prevention of outbreaks in some individuals, including those who are immunocompromised[16]. Such frequent dosing schedules are generally not ideal for compliance and are particularly unpopular in patients with HIV infection who may already be taking numerous other drugs prescribed for the management of their HIV infection or attendant sequelae.

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Valaciclovir (Valtrex™) is the L-valyl ester of aciclovir which, after oral administration and absorption, is rapidly and almost completely converted to aciclovir and L-valine, the naturally occurring essential amino acid. Valaciclovir achieves high aciclovir bioavailability, 3- to 5-fold higher than can be achieved with oral aciclovir[17,18]. The multiple dose pharmacokinetics of valaciclovir and aciclovir in patients with advanced HIV disease (CD4+ lymphocyte count <150 cells/mm^3) have been extensively studied and are virtually identical to those in healthy volunteers[19]. Thus, valaciclovir offers the potential for sustaining or enhancing the efficacy of aciclovir in the management of recurrent HSV infection but with less frequent dosing regimens.

This paper describes 2 randomized, double-blind studies of valaciclovir in patients in the intermediate stages of HIV infection and disease. The objective of each study was to identify suitable valaciclovir dosage regimens for the effective management of recurrent HSV infection in patients immunocompromised as a consequence of HIV infection. The studies evaluate valaciclovir as suppressive therapy (secondary prophylaxis) and acute (episodic) treatment for recurrent anogenital HSV outbreaks and are presented side by side since both therapeutic strategies may be applicable in practice. The studies were conducted in the period 1991 to 1996, before HAART was widely used. The benefit of HAART in reducing the frequency and severity of HIV-related opportunistic infections is well recognized and its relatively widespread use in western countries has even prompted reassessment of the role of prophylaxis[20]. However, some patients do not tolerate HAART nor cope with the complex drug regimens involved. Furthermore, immune reconstitution with HAART may only be partial, to levels still associated with an increased risk of latent HSV reactivation[20,21]. Hence, the management of recurrent HSV infections in HIV-infected subjects should be considered in the context of the overall strategy and effectiveness of HIV management.

MATERIALS AND METHODS

Study designs

Study 1 was a 3-arm, randomized, double-blind, parallel comparison of valaciclovir 500 mg twice daily, valaciclovir 1000 mg once daily or aciclovir 400 mg twice daily for the prevention (suppression) of recurrent episodes of anogenital HSV infection in HIV-infected individuals. Patients were randomized using a 1:1:1 allocation to treatment and received continuous study medication for a maximum period of 48 weeks.

Study 2 was a 2-arm, randomized, parallel comparison of patient-initiated therapy with valaciclovir 1000 mg twice daily and aciclovir 200 mg 5 times daily, for the episodic treatment of a single episode of recurrent anogenital HSV infection in HIV-infected patients. Patients were allocated 1:1 to treatment. Treatment was initially for 5 days, but if there was insufficient evidence of resolution of signs or symptoms after 5 days, patients were offered a further 5-day course of study medication.

Patient populations

Patients in both studies were HIV-seropositive, with a history of recurrent HSV infection in the genital or anogenital area. Patients were excluded if they had evidence of clinically significant renal or hepatic impairment (estimated creatinine clearance $\leqslant$35 ml/min or $\geqslant$3-fold increase in liver transaminase function), malabsorption syndrome or a history of hypersensitivity to aciclovir. Pregnant women or nursing mothers were also excluded. Antiretroviral therapy was not restricted. Studies were conducted between August 1991 and September 1996, during which period HAART was not widely used.

Patients eligible to participate in Study 1 (suppressive therapy) were HIV-infected adults of either sex aged $\geqslant$18 years with an absolute CD4+ lymphocyte count $\geqslant$100 cells/mm^3 at enrolment. A history of recurrent anogenital herpes with documented positive HSV culture within the previous 5 years was required, with evidence of one or more recurrences of anogenital HSV in the year prior to study entry. Co-medication with systemic antiherpes therapy or interferon was not allowed.

Eligible subjects for Study 2 (episodic treatment) were aged $\geqslant$18 years old and of either sex. There was no lower limit for absolute CD4+ lymphocyte count. Patients were required to have suffered 2 or more recurrent anogenital HSV episodes in the 6 months preceding study entry, or since stopping suppressive aciclovir therapy. Patients receiving systemic antiherpes treatment in the 7 days before starting study medication, or immunomodulatory treatments in the 30 days before, were not eligible.

Institutional Review Board approval was obtained at each study site before enrolment commenced. All subjects gave informed, written consent.

Study procedures

Study 1 (suppressive therapy)

Eligible patients were randomly assigned (1:1:1) to one of 3 treatments using a computer-generated randomization schedule: valaciclovir 500 mg twice daily or 1000 mg once daily or aciclovir 400 mg twice daily. The identity of the medication was detailed in sealed envelopes, which were only opened if considered clinically necessary by the investigator. Blood samples were obtained for baseline haematology (haemoglobin, total white cell count, differential neutrophil and lymphocyte counts, platelets) and biochemistry (creatinine, alkaline phosphatase, alanine or aspartate transaminase); a physical

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examination was performed and a brief HSV history taken. Study medication was supplied for 4-week periods, as valaciclovir 500 mg tablets or aciclovir 400 mg tablets each with matching placebo tablets. Patients were instructed to record times and dates of study drug administration, concomitant medications and any anogenital herpes episodes in a patient diary. Double-blind, double-dummy treatment started immediately on completion of screening and continued for 48 weeks.

Clinical evaluation was required every 4 weeks and patients additionally reported to the clinic within 24 h of the onset of any anogenital HSV recurrences. At each clinic visit, the patient diary was reviewed and the occurrence of any other herpes manifestation was documented. Adverse events reported by the patients were also noted. Blood samples were taken at weeks 24 and 48 for biochemistry and haematology and additionally at weeks 12 and 36 for serum creatinine. Plasma aciclovir concentrations were determined by radioimmunoassay from samples obtained at weeks 12, 24, 36 and 48[22].

When patients returned to the clinic with evidence of an anogenital HSV episode, lesions were staged, the extent of pain noted and a swab for HSV culture and aciclovir sensitivity testing requested. Following virus isolation and storage at −80°C, HSV sensitivity to aciclovir was determined using a plaque reduction assay in Vero cells, as previously described[23,24]. For patients with lesions at or beyond the macule/papule stage, episodic treatment with valaciclovir (1000 mg twice daily) was prescribed. After resolution of each recurrence, patients resumed suppressive therapy according to the original study randomization.

Study 2 (episodic treatment)

Patients were randomly allocated (1:1) to one of two 5-day treatment arms using a computer-generated randomization schedule: valaciclovir 1000 mg twice daily or aciclovir 200 mg 5 times daily. The identity of treatment was described in sealed envelopes which were only opened if considered clinically necessary by the investigator. Study medication was provided as capsules containing valaciclovir 250 mg or aciclovir 200 mg with respective matched placebo. Patients were instructed to self-initiate double-blind, double-dummy treatment within 24 h of the first signs or symptoms of their next anogenital HSV recurrence. A patient diary was used to record the date and time of the onset of symptoms, patient assessment of pain intensity and healing, and the exact times of study drug administration.

Patients returned for clinical evaluation within 24 h of starting treatment (day 1), at which time lesions were staged (macule/papule, vesicle/pustule/ulcer, crusted, healed) and swabs taken for aciclovir sensitivity testing by the method described above. Pain was recorded as present (mild, moderate, severe) or absent. Patients with insufficient improvement in cutaneous lesions or pain by day 5 were considered initial non-responders; such patients were offered additional study medication and evaluated until all signs and symptoms had resolved. Safety was evaluated from reports of adverse events during treatment and follow-up and from biochemistry and haematology on samples take on days 1 and 5.

Clinical endpoints

Study 1 (suppressive therapy)

The primary efficacy endpoint was the time from randomization to first recurrence of anogenital HSV infection. The time to second anogenital HSV recurrence was also recorded. In addition, the occurrence of non-genital HSV episodes, first evidence of other herpesvirus disease manifestations (including cytomegalovirus disease), non-herpesvirus opportunistic infections and death was also recorded.

Study 2 (episodic treatment)

The primary efficacy endpoint was episode duration, defined as the time from first dose to complete resolution of all symptoms and signs. This study prospectively identified 2 patient subgroups, according to clinical outcome, i.e. patients whose lesions 'aborted' and did not progress beyond the macule/papule stage and those whose lesions progressed through all stages before healing (complete re-epithelialization).

Statistical methods

Study 1 (suppressive therapy)

The sample size calculation was based on the analysis of time to first anogenital HSV recurrence using the log rank test[25]. Assuming that hazard functions were proportional, 300 patients per group provided 80% power to detect an improvement of 25% in the proportion of patients recurrence-free at one year (0.45 *vs* 0.56) using a 2-sided test at the 5% significance level. This treatment difference is equivalent to a hazard ratio of 0.72 or 1.39. Recruitment of 350 patients per group allowed for a 15% premature discontinuation rate.

Intent-to-treat comparison of each of the valaciclovir arms with aciclovir was of primary interest. The distribution of time to first recurrence was estimated using the Kaplan–Meier product limit method[25] and differences between treatments were estimated and tested using Cox's proportional hazards models[26]. The validity of the proportional hazards model (i.e. that the hazard ratio does not change with time) was checked by plotting the log of the negative log of the survival distribution against time. Study centre was fitted as a covariate into each Cox's model.

Herpes simplex virus sensitivity to aciclovir (Studies 1 and 2) was expressed in terms of IC_{50}

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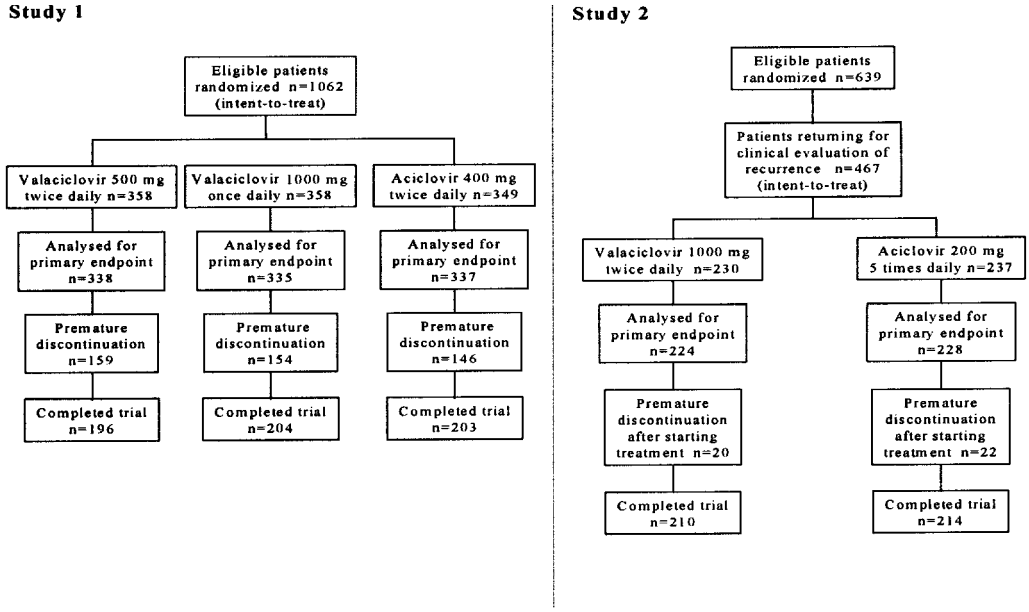


Figure 1. Progress of patients through the stages of Study 1 (suppressive therapy) and Study 2 (episodic treatment)

values, i.e. the concentration required to inhibit 50% of viral plaques compared with the control, and was derived using the Grafit software package[24].

Plasma aciclovir concentration–time data were analysed by population pharmacokinetic methods using nonlinear mixed effects modelling software (NONMEM version 4)[27]. Estimates of individual pharmacokinetic parameters were derived and descriptive statistics used to compare treatments.

Study 2 (episodic treatment)

The sample size estimate was based on the analysis of episode duration using the log rank test[25]. On the assumption that hazards functions were proportional, a minimum sample size of 450 patients per treatment group was sufficient to provide 80% power to detect a 15% reduction in the proportion of patients with residual signs or symptoms on day 5 of treatment at the 5% level of significance, assuming that 45% of the aciclovir group had signs or symptoms. This is equivalent to a hazard ratio of 0.83 or 1.20.

The intent-to-treat population was defined as all randomized patients returning to the clinic with a recurrence. The distributions of times to complete resolution of all signs and symptoms, and to lesion healing, were estimated using the Kaplan–Meier product limit method[25] and treatment differences tested using Cox's proportional hazards models[26], adjusting for any study centre effect. The validity of the proportional hazards model was checked by plotting the log of the negative log of the survival distribution against time.

Assessment of safety

Adverse events during treatment and follow-up were listed with the intensity of the event, its seriousness, investigator-attributed causality and duration. Numbers of patients reporting adverse events and the nature of events in each treatment group were compared. Hodges–Lehmann estimates and 95% CI were used to assess treatment differences or changes over time in haematology and biochemistry variables.

RESULTS

Study 1 (suppressive therapy)

A total of 1062 patients participated at 99 centres in Australia, Canada, Europe and the USA. Figure 1 shows the numbers of patients randomized to each treatment group and their progress through the study. There were no major differences in the demographic and disease characteristics between the treatment groups (Table 1). Most subjects were male (89–92%) and the median age was 36 years. The median CD4+ lymphocyte count at the start of therapy was 320 cells/mm^3 and 30% of the patients had signs of HIV disease. This study was completed prior to the era of quantitative HIV RNA measurement. The majority of subjects had suffered 2–5 anogenital HSV recurrences in the previous year; approximately half had received suppressive aciclovir therapy at some time during this period, prior to stopping it for participation in the present trial. Overall, 43% of the study population discontinued prematurely during the treatment period, mainly as a result of a subsequent personal decision not to participate further (16%) and loss to follow-up (13%). Other reasons were adverse experience (8%), protocol violation (6%) and death (<1%). Reasons for premature discontinuation were evenly distributed across treatments.

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Table 1. Demographic details and disease characteristics—Studies 1 and 2

	Study 1			Study 2	
	Valaciclovir 500 mg twice daily *n*=355	Valaciclovir 1000 mg once daily *n*=358	Aciclovir 400 mg twice daily *n*=349	Valaciclovir 1000 mg twice daily *n*=230	Aciclovir 200 mg 5 times daily *n*=237
Female/male (%)	10/90	8/92	11/89	7/93	11/89
Age (years): median (range)	36 (20–66)	36 (21–66)	36 (21–60)	36 (19–61)	36 (19–75)
HIV status:					
Asymptomatic/symptomatic (%)	72/28	68/32	71/29	54/46	56/43
CD4+ count (cells/mm^3)* at enrolment:					
Median (range)	309 (0–1440)	320 (7–1522)	340 (0–1260)	232*	247*
Years since first HSV episode: median (range)	4.41 (0–39.7)	4.98 (0–27.3)	4.85 (0–33.3)	4.7 (0–34.1)	4.1 (0.1–24.3)
Recurrences in previous year (%):					
1	17	18	18	7	7
2–5	55	61	56	80	76
⩾6	27	22	26	13	17
Use of suppressive aciclovir in previous year:					
No/Yes (%)	55/44	57/43	53/47	74/26	73/27

*Not specified in protocol: data available for 353 patients

Clinical outcome measures

Recurrent HSV outbreaks

There were no statistical differences between either valaciclovir regimen and aciclovir for the time to first anogenital HSV recurrence. For valaciclovir 500 mg twice daily *vs* aciclovir, the hazard ratio [CI] was 0.73 [0.50, 1.06], P=0.10, illustrating a trend towards the superiority of valaciclovir twice daily (Figure 2). For valaciclovir 1000 mg once daily *vs* aciclovir, the hazard ratio [CI] was 1.31 [0.94, 1.82], P=0.11, suggesting inferiority of once daily valaciclovir compared with twice daily aciclovir. The hazard ratio [CI] for valaciclovir 1000 mg once daily *vs* 500 mg twice daily of 1.80 [1.26, 2.57], P=0.001 substantiates the superiority of twice daily valaciclovir. Kaplan–Meier estimates of patients remaining recurrence-free at 48 weeks were 82%, 71% and 78% for valaciclovir twice daily, valaciclovir once daily and aciclovir, respectively.

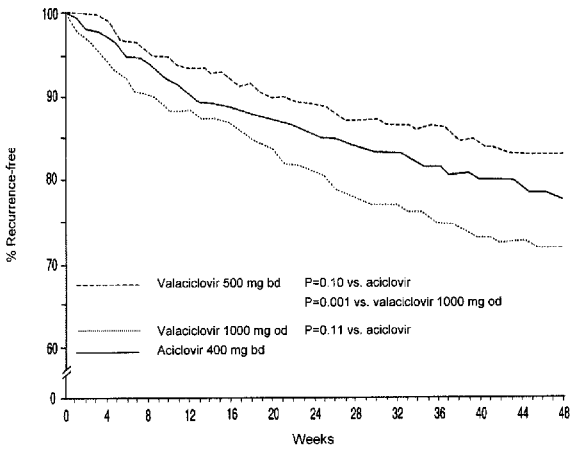


Figure 2. Cumulative proportions of patients recurrence-free during suppressive therapy (Study 1). bd=twice daily; od=once daily

Comparisons of each valaciclovir regimen with aciclovir on time to second anogenital HSV recurrence strongly reflected the results of the intent-to-treat comparisons for the primary endpoint. Hazard ratios [CI] for valaciclovir 500 mg twice daily and 1000 mg once daily *vs* aciclovir were 0.89 [0.44, 1.79] and 1.69 [0.92, 3.13], respectively. For the comparison of once *vs* twice daily valaciclovir, the hazard ratio [CI] was 1.90 [1.03, 3.52]. From confidence intervals, the difference between valaciclovir 500 mg twice daily and 1000 mg once daily was significant.

Other herpesvirus infections

In total, 85 (8%) individuals experienced one or more HSV recurrences at a non-genital site. Eleven patients suffered one or more recurrences of cytomegalovirus disease, 3 in the valaciclovir 500 mg twice daily group and 4 each in the valaciclovir 1000 mg once daily and aciclovir groups. There were no reports of Epstein–Barr-associated lymphoma or lymphoproliferative disease, nor of varicella zoster virus reactivation during the study period. Kaposi's sarcoma was reported in 44 subjects overall, 9 and 22 cases, respectively, in the twice and once daily valaciclovir groups and 13 in those receiving aciclovir. In view of the low incidence of clinical endpoints for conditions other than anogenital HSV, statistical hypothesis testing for treatment differences was considered inappropriate.

Aciclovir sensitivity of breakthrough HSV isolates

Two viable HSV isolates were obtained for aciclovir sensitivity testing, from 2 subjects who experienced a breakthrough episode after 26 and 30 weeks of valaciclovir 1000 mg once daily. IC_{50} values were 2.3 and 2.5 μM, respectively [control range 0.95–3.9 μM].

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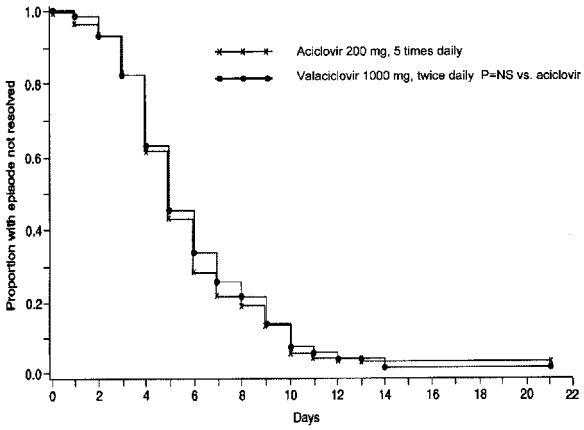


Figure 3. Cumulative proportions of patients with signs or symptoms of an anogenital herpes simplex virus recurrence (Study 2). NS=not statistically significant

Plasma aciclovir level monitoring

Estimated mean peak (range) aciclovir concentrations for patients allocated valaciclovir 500 mg twice daily, valaciclovir 1000 mg once daily and aciclovir 400 mg twice daily were 9.5 (5.6–24.5), 17.2 (7.9–34.4) and 2.6 (1.4–9.6) μM, respectively. Mean trough (range) aciclovir concentrations were 1.24 (0–8.4), 0.27 (0–6.5) and 0.58 (0.09–7.9) μM for the twice and once daily valaciclovir and aciclovir regimens, respectively. Corresponding estimates of the mean area under the curve (AUC_{0-24}) were 108.8 (48.0–318.3), 104.3 (46.6–293.9) and 37.3 (19.5–216.2) μM.h, respectively. Although peak concentrations increased approximately 3.7- and 6.6-fold, respectively with the valaciclovir twice daily and once daily regimens over aciclovir, the systemic aciclovir exposure estimated from AUC_{0-24} increased by only 3-fold and was similar for both valaciclovir regimens.

Study 2 (episodic treatment)

A total of 639 patients were recruited in 77 centres in Australia, Europe and the USA. The intent-to-treat population comprised 467 patients at 56 centres who returned for clinical evaluation of a recurrence prior to study termination. Figure 1 summarizes the numbers of patients randomized, those returning for clinical evaluation and their progress through the study. The demographic data and disease characteristics indicated no major differences between the treatment groups (Table 1). Most participants were male (89–93%) and the median age was 36 years. Almost half of the patients (43–46%) had clinical signs of HIV disease.

Clinical outcome measures

HSV episode duration

No differences were detected between the valaciclovir 1000 mg twice daily and aciclovir 200 mg 5 times daily regimens on episode duration or lesion healing (Figure 3). The hazard ratio [CI] for valaciclovir *vs* aciclovir on episode duration was 0.93 [0.75, 1.14], P=0.45, and the median episode duration was 5 days. Lesions were observed to progress through all stages to healing in 394 of the 467 treated patients. Data on the remaining patients were incomplete, therefore lesions could not be assumed to have 'aborted'. The hazard ratio [CI] for valaciclovir *vs* aciclovir on lesion healing was 0.98 [0.79, 1.22], P=0.89, and the median healing time was 5 days. In this study, only the primary efficacy endpoints were analysed because of premature study closure due to slow recruitment rates.

Aciclovir sensitivity

Herpes simplex virus isolates were obtained from only 8 subjects on days 1, 2 or 3 of episodic treatment with valaciclovir (n=4) or aciclovir (n=7). Wild type sensitivity to aciclovir was demonstrated for each isolate, with IC_{50} values ranging from 1.0 to 2.5 μM [control range 0.95–3.9 μM].

Table 2. Most common adverse events (% patients reporting)

	Study 1			Study 2	
	Valaciclovir 500 mg twice daily n=355	Valaciclovir 1000 mg once daily n=358	Aciclovir 400 mg twice daily n=349	Valaciclovir 1000 mg twice daily n=230	Aciclovir 200 mg 5 times daily n=237
Diarrhoea	19	21	19	7	9
Headache	18	18	17	8	8
Infection	16	19	13	1	1
Rash	14	16	14	1	1
Nausea	16	13	12	10	5
Rhinitis	13	12	14	0	1
Pharyngitis	11	11	13	1	1
Abdominal pain	12	10	7	3	3
Fever	11	10	11	1	3
Depression	9	11	7	0	0
Cough	6	10	10	0	0

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Analysis of safety

Study 1 (suppressive therapy)

The nature and frequency of adverse events were similar in patients receiving continuous valaciclovir or aciclovir for prevention of anogenital HSV recurrences (Table 2); headache and nausea were most frequent. Most reported events were not considered attributable to study medication. Overall, 11% of valaciclovir recipients and 9% of aciclovir recipients experienced treatment-limiting adverse events, most commonly nausea, headache and diarrhoea. No treatment-related differences in haematology or biochemistry, or changes with time were evident. Median serum creatinine values remained constant within the range of 88–93 μM at each quarterly evaluation.

The distribution of opportunistic infections typical of HIV infection and disease across the treatment groups was similar. In total, 1156 HIV-associated (non-herpesvirus) opportunistic infections were reported in 485 patients: fungal infections were the most common, occurring in 21% of patients. *Pneumocystis carinii* pneumonia developed in 18 participants. There was one case each of non-Hodgkin's lymphoma (valaciclovir 500 mg twice daily) and toxoplasmosis (valaciclovir 1000 mg once daily). None of the opportunistic infections was considered attributable to valaciclovir or aciclovir administration.

Ten deaths were recorded during the treatment period; 3 were in valaciclovir 500 mg twice daily recipients, 5 in valaciclovir once daily and 2 in aciclovir recipients. One death was considered possibly attributable to study treatment, a case of lactic acidosis in an aciclovir recipient who was also receiving numerous other medications including stavudine, omeprazole, cisapride and fluconazole.

Study 2 (episodic treatment)

Adverse events during treatment were generally mild in intensity. Headache, nausea and diarrhoea were the most commonly reported and occurred in $\leqslant$10% of subjects (Table 2). Adverse events resulting in premature discontinuation of treatment occurred in 7 individuals, 3 randomized to valaciclovir (one case each of depression/alcohol abuse, vasodilatation/headache/meningitis/nausea and abdominal pain/rash) and 4 allocated to aciclovir (one case each of headache/diarrhoea/fever, erythema multiforme, rash and pancreatitis/hepatitis). There were no clinically meaningful changes over time in haematology or biochemistry nor differences between treatment groups.

DISCUSSION

These 2 studies demonstrate the clinical utility of valaciclovir and oral aciclovir for management of recurrent anogenital HSV infections in immunocompromised, HIV-infected patients. When used continuously, valaciclovir suppressed recurrent HSV outbreaks. When used to treat individual HSV episodes, twice daily valaciclovir was as effective as 5 times daily oral aciclovir in terms of speeding lesion healing and alleviating pain and discomfort. Although the studies were not designed nor powered for demonstrating the statistical equivalence of the respective valaciclovir and aciclovir regimens, the findings can be considered as indicative of clinical equivalence. The safety profile of valaciclovir was not appreciably different from that of aciclovir, although as expected, many opportunistic infections were also reported during the one-year study; these were generally not different from those typically seen in subjects with HIV infection and were not considered a consequence of antiherpes therapy.

Valaciclovir was developed as an oral pro-drug of aciclovir to improve its bioavailability, thus reducing the need for frequent dosing regimens. In previous studies of valaciclovir for the acute treatment of initial and recurrent genital herpes in immunocompetent subjects, the efficacy of twice daily valaciclovir was comparable with that of 5 times daily aciclovir[28,29]. Furthermore, valaciclovir 500 mg twice daily was as effective as 1000 mg twice daily[30]. It remains to be established if twice daily valaciclovir 500 mg will be as effective as 1000 mg twice daily for the episodic treatment of genital herpes in HIV-seropositive individuals. Effective suppression of recurrent HSV episodes is achieved with once daily valaciclovir in immunocompetent individuals. However, for patients with more frequent or severe HSV outbreaks, a higher or divided dose may be more beneficial[31]. This is logical and consistent with previous studies of aciclovir in immunocompetent and immunocompromised subjects in which higher unit doses given more frequently provided more effective control of recurrent HSV outbreaks[16,32]. Our results are supportive of earlier observations in those at risk of severe HSV outbreaks, in that better suppressive efficacy was evident with valaciclovir 500 mg twice daily compared with 1000 mg once daily in immunocompromised persons with HIV infection.

Patients receiving continuous suppressive therapy in Study 1 had less advanced HIV disease than those treating an individual HSV recurrence (Study 2, episodic treatment). This was a design feature of Study 1 in an attempt to minimize patient drop-out during the one-year treatment period. Although suppressive valaciclovir 500 mg twice daily was not statistically superior to the widely-used standard aciclovir regimen, our data suggest that this valaciclovir regimen offered better protection than aciclovir. The 43% premature discontinuation rate in Study 1 (range 42–45% across groups), together with an overall low anogenital HSV recurrence rate, undoubtedly compromised its statistical power. Nevertheless, 27% prevention or delay in time to first HSV recurrence afforded by

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twice daily valaciclovir compared with aciclovir was close to the 28% estimate (hazard ratio 0.72) used for the sample size calculation.

For immunocompromised HIV-infected subjects unable to mount an adequate immune response to a HSV reactivation episode, it may be important to maintain trough aciclovir plasma concentrations above a critical level in relation to the *in vitro* IC_{50} value for HSV to ensure that inhibitory concentrations are sustained throughout most of the day. The present study demonstrated that the valaciclovir 500 mg twice daily regimen gave trough plasma aciclovir levels that were almost 5 times greater than those achieved on the valaciclovir 1000 mg once daily regimen (1.24 *vs* 0.27 μM) and 2-fold greater than after oral aciclovir. Of note, aciclovir peak levels and AUC_{0-24} values did not correlate with outcome.

The few viable HSV isolates obtained in these 2 studies mean that the aciclovir susceptibility results should be considered anecdotal. All isolates demonstrated wild type sensitivity to aciclovir. Data are accumulating on HSV susceptibility to antivirals in HIV-seropositive subjects with some evidence that resistance to aciclovir occurs, albeit rarely[10,33]. The data from Study 2 (episodic treatment) provide a useful estimate of 'baseline' sensitivity in this immunocompromised population and indicate that prevalent HSV strains will be responsive to valaciclovir treatment; furthermore, they are likely to remain so with chronic suppressive therapy.

Recruitment into Study 2 (episodic treatment) was terminated prematurely due to slow recruitment associated with eligible patients' preference being for suppressive therapy at the time studies were conducted. This reduced the power to detect statistically significant differences between valaciclovir and aciclovir episodic treatment. However, from the available data, the rates of episode resolution and lesion healing were within a 10% range indicative of clinical equivalence. The twice daily valaciclovir regimen will be more convenient than 5 times daily aciclovir for HIV-infected persons, who already experience a substantial 'pill burden'.

There was no evidence in either the episodic or continuous therapy studies of any increased incidence of adverse events with valaciclovir treatment, in particular the thrombotic microangiopathy (TMA)-like syndrome. Most adverse events were not considered attributable to study medication, but rather to the characteristics of HIV infection and disease. For Study 1 (suppressive therapy), the incidence of adverse events appears lower than in comparable studies in immunocompetent patients[31,34]. The tendency towards underreporting may be due to many events being considered typical of underlying HIV disease and did not merit specific recording.

Thrombotic microangiopathy-like syndrome has been observed in severely immunocompromised subjects with advanced HIV disease in a study investigating valaciclovir as prophylaxis for cytomegalovirus infections and disease[35,36]. Higher dosages of 8 g valaciclovir daily were administered in this trial, increasing systemic exposure to aciclovir by approximately 7-fold. Any relationship between TMA-like syndrome and valaciclovir (and aciclovir) remains to be fully elucidated, but it is noteworthy that the syndrome is also correlated with advanced HIV disease and other medical conditions, including severe iatrogenic immunosuppression and other classes of drug, particularly antifungal agents[35–38]. The absence of any symptoms suggesting TMA-like syndrome in the 1062 participants of our one-year suppression study is consistent with current thinking that the occurrence of TMA in valaciclovir clinical trials is restricted to severely immunosuppressed individuals receiving higher dosages than are required to control HSV infection.

The results of our 2 studies with valaciclovir are encouraging and offer improved options for management of recurrent HSV infection in immunocompromised individuals. Whilst the advent of HAART for the management of HIV infection may encourage consideration of a switch from continuous suppressive therapy towards episodic treatment of individual herpes outbreaks, there are no data to suggest a decrease in HSV shedding while on HAART, especially among persons with a continued low CD4+ count[9,10]. Whether suppressive therapy is the ideal option for HSV disease management in subjects with HIV co-infection is unknown. The decision must ultimately balance the potential benefits of durable HSV suppression with other factors such as cost, morbidity and the effects, if any, of chronic drug exposure.

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